Single-phase Boost PFC Converter - II

Introduction

The rapid expansion of electric vehicles and other modern electronics necessitates robust and efficient charging infrastructure. A critical component in this infrastructure is the Power Factor Correction (PFC) converter, especially in single-phase applications. This lecture delves into the intricacies of the single-phase boost PFC converter, a topology widely used to ensure that power drawn from the AC mains is clean, efficient, and compliant with grid standards. Understanding this converter is vital for engineers designing power supplies that not only deliver stable DC output but also minimize harmonic distortion and reactive power consumption, thereby contributing to a more stable and efficient electrical grid. This session builds upon previous discussions, focusing on how the converter achieves its dual objectives: maintaining a constant output DC voltage and drawing a sinusoidal input current in phase with the input voltage, thus achieving unity power factor.

Core Concepts

Term	Definition	Significance
`Boost PFC Converter`	A power electronic circuit combining a full-bridge diode rectifier with a boost converter, designed to improve power factor and regulate output voltage.	Essential for drawing sinusoidal current from the AC source and providing a stable DC output for loads like EV chargers.
`Unity Power Factor (UPF)`	A condition where the input current drawn from the AC source is perfectly in phase with the input voltage, minimizing reactive power.	Crucial for reducing energy losses, improving grid efficiency, and complying with power quality standards.
`Rectified DC Voltage (vd)`	The output voltage of the full-bridge diode rectifier, which is the absolute value of the input AC voltage (e.g.,	Vs,pk sin(ωt)
	The fraction of a switching period during which a switch	The primary control variable that shapes the input current to

Term	Definition	Significance
`Duty Ratio (D(t))`	(S1) is turned on, dynamically varied over time.	achieve unity power factor and regulates the output voltage.
`Volt-second Balance`	The principle stating that the average voltage across an inductor over a complete switching cycle must be zero in steady state.	Used to derive the relationship between input and output voltages and the duty ratio, crucial for converter design.
`Switching Frequency (fsw)`	The frequency at which the converter's switches (S1, S2) are turned on and off, typically in the tens of kilohertz range.	A high switching frequency allows for smaller inductor and capacitor sizes and enables the approximation for volt-second balance.
`Ripple Current (ΔiL)`	The peak-to-peak variation in the inductor current within a single switching cycle.	A key design parameter for the inductor, influencing its size, losses, and the overall current quality.
`Vs,peak`	The peak amplitude of the input AC voltage.	Defines the maximum instantaneous input voltage, which is a critical reference for determining the minimum output voltage requirement of the boost converter.

Detailed Analysis

1. Operating Principles and Control Objectives

The single-phase boost PFC converter integrates a full-bridge diode rectifier with a boost converter. The rectifier converts the AC input voltage ($V_s = V_{s,pk} \sin(\omega t)$) into a rectified DC voltage ($v_d = |V_{s,pk} \sin(\omega t)|$). This rectified voltage, which varies with the modulus of the sine wave, then feeds into the boost converter stage. The primary goal of this boost stage is twofold:
* **Unity Power Factor (UPF):** To ensure that the current drawn from the AC source (i_s) is sinusoidal and perfectly in phase with the input voltage (V_s). This implies that the average inductor current ($i_{d,average}$) must follow the same rectified sinusoidal pattern as v_d.

* **Constant Output Voltage (Vo):** To maintain a stable DC voltage at the output, despite the fluctuating input voltage. This output voltage must be regulated within permissible limits for the connected load.

To achieve these objectives, the boost converter's switches (S1 and S2) are controlled dynamically. The switching frequency (fsw) is significantly higher than the supply frequency (fs), typically tens of kilohertz compared to 50 Hz. This high switching frequency is a critical assumption that allows for certain approximations in the analysis, particularly regarding the volt-second balance across the inductor. The control strategy involves varying the turn-on times of S1 and S2 such that the average current drawn from the rectifier (id) mirrors the shape of vd, while simultaneously regulating Vo.

The inductor current (iL, which is the same as id) exhibits a ripple around its average value. Over a half-cycle of the AC input, the average inductor current starts from zero, rises to a peak, and then returns to zero, following the shape of vd. The actual current waveform consists of these average variations superimposed with high-frequency ripples due to switching. Increasing the switching frequency reduces the ripple magnitude and makes the current waveform appear more sinusoidal, closer to the desired UPF operation.

2. Duty Ratio Derivation for PFC and Voltage Regulation

The duty ratio, D(t), is the fraction of the switching period (Ts) during which the main switch (S1) is turned on. This duty ratio must vary continuously over the AC half-cycle to satisfy both the UPF and constant output voltage requirements. To derive D(t), we apply the principle of volt-second balance across the inductor (L).

In a boost converter, during the ON-time (D(t) * Ts) of switch S1, the voltage across the inductor is approximately equal to the input voltage, vd(t) = |Vs(t)|. During the OFF-time ((1 - D(t)) * Ts), the voltage across the inductor is vd(t) - Vo.
Traditionally, for DC-DC converters, volt-second balance is applied over a single switching cycle, assuming the inductor current returns to its starting value.
However, in a PFC converter, the average inductor current varies over the AC half-cycle. The lecture makes a crucial approximation: since fsw is much greater than fs, the change in the average current over one switching period is very small.
Therefore, for practical purposes, we can assume that the inductor current at the beginning and end of a switching cycle is approximately the same, allowing us to apply the volt-second balance over a single switching cycle.

Applying volt-second balance:
(vd(t) * D(t) * Ts) + ((vd(t) - Vo) * (1 - D(t)) * Ts) = 0

Dividing by Ts and expanding:
vd(t) * D(t) + vd(t) - vd(t) * D(t) - Vo + Vo * D(t) = 0

$vd(t) - Vo + Vo * D(t) = 0$
$Vo * D(t) = Vo - vd(t)$
$D(t) = (Vo - vd(t)) / Vo$
$D(t) = 1 - (vd(t) / Vo)$

Substituting $vd(t) = |Vs,pk \sin(\omega t)|$:
$D(t) = 1 - (|Vs,pk \sin(\omega t)| / Vo)$

This expression shows that the duty ratio is not constant but varies with the instantaneous input voltage.
A critical condition arises from this derivation: for D(t) to be a valid duty ratio (i.e., between 0 and 1), Vo must always be greater than Vs,pk. If Vo were less than Vs,pk, D(t) could become negative, which is physically impossible as time cannot be negative. This inherent characteristic explains why it's called a "boost" PFC converter; it always boosts the output voltage above the peak input voltage. For example, if the input is 230 V RMS (Vs,pk ≈ 325 V), the output DC voltage (Vo) must be greater than 325 V, typically chosen around 400 V to account for voltage drops and provide sufficient boosting margin. The duty ratio waveform will be the inverse of the rectified input voltage waveform, starting at 1 (when vd is 0), decreasing to a minimum value (1 - Vs,pk/Vo) when vd is at its peak, and then increasing back to 1.

3. Inductor Sizing and Ripple Current Analysis

Sizing the inductor (L) is crucial for managing the ripple current (ΔiL) and ensuring continuous conduction mode. The ripple current is defined as the difference between the maximum and minimum inductor current within a switching cycle ($\Delta iL = iL,max - iL,min$).

During the ON-time ($D(t) * Ts$), the voltage across the inductor is vd(t). The change in inductor current is given by:
$\Delta iL = (vd(t) / L) * D(t) * Ts$ (Equation 1)

During the OFF-time ($(1 - D(t)) * Ts$), the voltage across the inductor is $vd(t) - Vo$. The change in inductor current (magnitude) is:
$\Delta iL = (|vd(t) - Vo| / L) * (1 - D(t)) * Ts$ (Equation 2)
Since $Vo > vd(t)$, the term $(vd(t) - Vo)$ is negative, indicating a negative slope for the current. We use the magnitude for ΔiL.

From Equation 1, we can express $D(t) * Ts$:
$D(t) * Ts = (L * \Delta iL) / vd(t)$

From Equation 2, we can express $(1 - D(t)) * Ts$:
$(1 - D(t)) * Ts = (L * \Delta iL) / (Vo - vd(t))$

Summing these two time intervals gives the total switching period Ts:
$Ts = (L * \Delta iL) / vd(t) + (L * \Delta iL) / (Vo - vd(t))$

Ts = L * ΔiL * [(Vo - vd(t) + vd(t)) / (vd(t) * (Vo - vd(t)))]
Ts = L * ΔiL * [Vo / (vd(t) * (Vo - vd(t)))]

Rearranging to solve for ΔiL:
ΔiL = (Ts / L) * (vd(t) * (Vo - vd(t))) / Vo
Since Ts = 1 / fsw:
ΔiL = (1 / (L * fsw)) * (vd(t) * (Vo - vd(t))) / Vo
ΔiL = (vd(t) / (L * fsw)) * (1 - (vd(t) / Vo))

To find the maximum ripple current (ΔiL,max), we differentiate ΔiL with respect to vd(t) and set the derivative to zero.
Let x = vd(t). We want to maximize f(x) = x * (1 - x/Vo) = x - x^2/Vo.
d(f(x))/dx = 1 - (2x / Vo) = 0
1 = 2x / Vo
x = Vo / 2

So, the maximum ripple current occurs when the instantaneous input voltage vd(t) is equal to half of the output voltage (Vo/2).
Substituting vd(t) = Vo/2 back into the ΔiL equation:
ΔiL,max = ((Vo/2) / (L * fsw)) * (1 - ((Vo/2) / Vo))
ΔiL,max = ((Vo/2) / (L * fsw)) * (1 - 1/2)
ΔiL,max = ((Vo/2) / (L * fsw)) * (1/2)
ΔiL,max = Vo / (4 * L * fsw)

Finally, rearranging to solve for the inductance L:
L = Vo / (4 * ΔiL,max * fsw)

This formula allows designers to calculate the required inductance. Vo is a specified output voltage, fsw is chosen by the designer (considering losses and EMI), and ΔiL,max is typically specified as a percentage of the peak input current (e.g., 5-10%). This ensures that the inductor is appropriately sized to limit current ripple to an acceptable level, preventing excessive losses and ensuring stable operation.

Key Takeaways

- [ ] The single-phase boost PFC converter uses a full-bridge rectifier followed by a boost stage to achieve both unity power factor and constant output DC voltage.
- [ ] The input current (id) must follow the rectified input voltage (vd) waveform to achieve unity power factor.
- [ ] The duty ratio D(t) is not constant but varies dynamically with the instantaneous input voltage, calculated as D(t) = 1 - (vd(t) / Vo).
- [ ] For proper operation, the output DC voltage (Vo) must always be greater than the peak input AC voltage (Vs,pk).

- [ ] The assumption of switching frequency (fsw) being much greater than supply frequency (fs) is crucial for approximating volt-second balance over a single switching cycle.
- [ ] The maximum inductor ripple current (ΔiL,max) occurs when the instantaneous input voltage vd(t) is equal to half of the output voltage (Vo/2).
- [ ] The inductor value (L) is determined by the output voltage, maximum allowable ripple current, and switching frequency using the formula L = Vo / (4 * ΔiL,max * fsw).
- [ ] Inductor sizing involves a trade-off between ripple current magnitude, inductor size, and switching frequency.

Conclusion

This lecture has provided a comprehensive understanding of the operational principles and design considerations for the single-phase boost PFC converter. By dynamically adjusting the duty ratio, this converter effectively shapes the input current to achieve unity power factor, thereby minimizing reactive power and harmonic distortion, which are critical for grid compatibility and energy efficiency. Simultaneously, it maintains a stable DC output voltage, essential for reliable operation of connected loads such as electric vehicle chargers. The derivation of the time-varying duty ratio and the detailed analysis of inductor sizing, including the calculation of maximum ripple current, are fundamental steps in designing a robust and efficient PFC stage. Future lectures will build upon this foundation by exploring the sizing of other critical components like the output capacitor and analyzing the voltage and current ratings of switches and diodes, ultimately leading to a complete understanding of the converter's control strategies for optimal performance.